Plan: from the density burst to typed vessel junctions

Sections 19–27 of the burst notebook · builds on vascnet · October 2026

This plan continues the burst notebook (sections 1–18) with sections 19–27. It follows the notes at the end of section 18:

- centerlines come from orientation-sensing neurons, not binary masks;
- lateral inhibition and a contrast layer do the thinning;
- a second, flow-sensitive field runs on frame differences;
- pruning is parametric;
- static and flow centerlines are compared and combined.

It then carries the result through to junction classification.

Principles

- **No binary masks for structure.** Centerlines are crests of an orientation field. They are thinned by lateral inhibition and strung together in position and orientation. Masks are only for display.
- **Nothing is trained.** Kernels are fixed. Thresholds are multiples of the background noise measured on this burst. Sizes are in pixels.
- **Two fields, one mechanism.** The static field sees density; the flow field sees how density changes. Both use the same hypercolumns, the same lateral inhibition and the same stringing.
- **Every step ends with a check.** That means a picture and a number: repeatability across fixations, loose ends per 1000 px, or truth on a synthetic burst.
- **Reuse vascnet where it already does the job.** That covers kernels, orientation score, association field, orientation-space thinning, junction rules, synthetic data and metrics.

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0. Small changes upstream

1. **Frame times.** `gate_burst_frames` computes `keep_indices` but does not return it. Return it as `gated_indices`, and add `frame_times = gated_indices / fps` and `fixation["times"]`. The flow field divides by the real time between kept frames, and gating leaves gaps.
2. **Data area per fixation.** `fixation_has_data[n]` is where `density_means[n]` is finite and `background_fits[n]["ok"]` holds, shrunk by `EDGE_MARGIN`.
3. **One temporal frame.** `temporal_frame(n, i) = density_fixations[n][i] - enlarge(frame_shadings[n][i])`. This is a frame's density with its moving shading taken out, positive in vessels. All flow work starts from it.
4. **vascnet accepts this notebook's density.**
   - Use `VesselPipeline.from_od(od)` with `od` = this notebook's density. The conventions match: `B - ln I`, positive in vessels.

- Add a `valid` mask to `cortex.feedforward`. The noise level and firing threshold are measured only on valid pixels, and responses within one kernel radius of no-data are zeroed.
- Set `radii ≈ RIDGE_WIDTHS / 0.836`. vascnet's radius is the cylinder radius; section 18's widths are Gaussian σ.

Phase A — the static orientation field

19. Contrast layer

These are the contrast-sensing neurons the section-18 notes ask for. Their output is "how many times the local noise this pixel stands above the background".

- **Input:** `background_fits[n]["OD"]` (alternative: `density_means[n]`).
- **No data:** continue the picture smoothly from around the hole, as `ridge_strength` does, with 0 (background density) far from data.
- **Normalize:** `contrast = OD / background_fits[n]["noise"]`. Density has already divided out the illumination; this divides out the local noise. A faint vessel in a dim corner and a dark vessel in the centre are then judged on the same scale.
- **Result:** `contrast_maps[n]`, `valid_maps[n]`.
- **Check:** on quiet pixels, `contrast` is roughly zero-mean with a robust sd within 10% of 1.

20. Orientation hypercolumns

- **Channels:** 16 orientations × half-widths `σ_v ∈ RIDGE_WIDTHS[:8]`. Scales of 6 px or more are computed on 2× or 4× reduced grids, as section 18 does.
- **Three receptive fields per channel** (`vascnet.kernels.line_pair`, elongation 2.5): even (line), odd (edge) and front half of the even kernel (end-stopping).
- **Complex unit:** `L = [E − α|O| − β|E_front − E_back|]₊` with α = β = 1. The odd term rejects edges. The front/back term rejects junction blobs and line ends, which is what keeps spurs from forming at every junction. It was the single most important component in vascnet's ablations.
- **Scale competition:** the strongest scale wins, refined by a parabola in log σ. The response is expressed in noise units, then re-referenced to the background floor of the strongest channel.
- **Result:**
  - `orientation_scores[n]`: (16, rows, columns) in noise units.
  - `radius_maps[n]`: (16, rows, columns).
  - One fixation at a time, float16 if needed. 16 × 2048² × 4 B is about 270 MB.
- **Check:**
  - On the large vessels, the winning orientation agrees with the structure-tensor orientation.
  - The radius map agrees with the half-widths section 18 found.

21. Orientation-domain interaction and lateral inhibition

This is the step that makes centerlines. It applies three fixed interactions, `LATERAL_STEPS` (3–5) times, each as a relaxed update `U ← (1 − η)·U + η·f(U)`.

1. **Tuning in orientation** ("a Gaussian in the orientation domain with side lobes suppressed"). Convolve each pixel's 16-vector over θ with a difference of Gaussians: centre σ_θ ≈ 1 bin, surround ≈ 3 bins, circular. This sharpens tuning and silences blobs that drive every orientation.
2. **Lateral inhibition across the vessel.** Channel θ is divided by its own activity on the flanks: `U_θ ← U_θ / (1 + k · (flank_θ ∗ U_θ))`. The flank kernel is two elongated Gaussian lobes at ±(1–2)·radius across the vessel. Iterating pulls a ridge profile in towards its crest and suppresses parallel side ridges.
3. **Facilitation along the vessel, modulatory only.** `U_θ ← U_θ · (1 + a·F/(h + F))`, with F the co-circular association field (`vascnet.cortex.AssociationField`). Continuity is strengthened, but nothing is drawn where there was no input. Additive "completion" made vascnet worse, so it stays off.

Protect crossings. Compute a co-occurrence signal `X = Σ √(U_θ · U_θ')` over channel pairs at least 3 bins apart, the analogue of V1's "plaid" cells. Where X is high, weaken the cross-orientation part of interaction 1. Otherwise a thin vessel crossing a thick one is erased inside the overlap, which turns the crossing into two T's.

Readout:

- **Crest points:** U_θ is largest across its own normal (orientation-space thinning, as in `vascnet.centerline.orientation_nms`).
- **Thresholds:** above `T_LOW` , and 8-connected within the channel to a point above `T_HIGH` .
- **Each crest point carries:** (x, y, θ, strength, radius).

Check:

- Cross-section profiles of 5 vessels before and after: the width at half maximum should fall towards 1–2 px.
- Count the parallel duplicate crests.

22. String the centerlines in position and orientation, without a skeleton

Stringing: from every crest point, step one pixel along ±θ. Accept the crest point found within 1 px of the predicted position whose orientation is within 1 bin, so a line may turn by up to one bin per pixel. Chains of accepted links are centerlines: polylines carrying θ, radius and strength at every point.

Crossings: two chains pass through the same pixel in different channels and stay two chains. A skeleton cannot do this.

Bridging gaps: join two chain ends that are at most `GAP_MAX` ·radius apart, if all of these hold:

- they can be joined by a smooth, monotonic curve turning no more than 90° (Kellman–Shipley relatability);
- the oriented evidence along the bridge exceeds `T_LOW / 2` ;
- for an end that abuts a wider vessel, the bridge is allowed only when the end sits flush against that vessel's wall.

Junction candidates:

- chain ends that come within (radius + 2) px of another chain;
- points where chains of different orientation meet.

Parametric pruning:

- **Pruned:** chains shorter than `L_MIN` or with mean strength below `Z_MIN` .
- **Sweep:** `L_MIN ∈ {5, 10, 20, 40}` px × `Z_MIN ∈ {2, 3, 4, 5}` × `T_HIGH ∈ {3, 4.5, 6}` .
- **Recorded for every setting:**
  - total length, segments and junction candidates;

  - loose ends per 1000 px;
  - share of length in the largest network;
  - found again in the other fixations (`found_again` from section 18).

  Plot each against the settings and pick a setting on a plateau, where small changes don't move the results. Don't pick the best-looking one.
- **Result:** `static_chains[n]` (list of polylines with per-point θ, radius, strength) and `pruning_table`.

Check: add these chains as a row of section 18's `graph_table`. Expect fewer loose ends than the mask-based graphs, and crossings that come out as two chains.

Phase B — the flow field

23. Is the blood's motion visible at this frame rate?

Settle this before building velocity. On 20–50 well-sampled points along the largest static chains:

- r_1**:** correlation between consecutive kept frames of `temporal_frame` minus its mean over time, at the same pixel.
- **Travelling correlation:** correlation between the line mass at chain position s in frame t and at s + Δ in frame t + 1, for Δ from −40 to +40 px along the chain. (Line mass is density integrated across the vessel; see 25a.)
  - A peak away from Δ = 0 means the pattern is visible flow moving Δ px per frame.
  - A flat curve near 0 means the pattern decorrelates within one frame, so tracking is aliased at this frame rate.
- **Report** by vessel width class. Also count the frame pairs split by gated-out frames. Derivative features only use pairs with `Δt ≤ MAX_GAP_FRAMES / fps`.

Decision:

- **Peaks found:** build velocity from kymographs (25b) for those width classes.
- **No peaks:**
  - Detect flow from decorrelation (section 24).
  - Take timing from slow signals (25c).
  - The flux test in phase D then uses slow or pulsatile flux only. Record the frame rate that would be needed, `fps ≥ v / (½ · pattern correlation length)`, for a later faster burst.

24. Flow-sensitive hypercolumns

This is the second neural field from the section-18 notes.

- **Input per kept pair (t, t+1) in a fixation:** `D = (temporal_frame(t+1) − temporal_frame(t)) / Δt`. Holes are filled with 0, and D is expressed in its own noise units, measured on quiet pixels.
- **Same hypercolumns as 19–21, with one difference:** D changes sign as red-cell clusters pass, so use the phase-invariant energy `E² + O²` of the even/odd pair (a complex cell), not the signed even response.
- **Don't store per-frame hypercolumns.** 16 × frames × rows × columns is tens of GB. Stream instead, and keep per fixation:
  - the time-averaged energy per orientation, `flow_energy[n]` (16, rows, columns), optionally lucky-weighted (25d);

    - the per-frame values sampled along static chains only, for section 25.
- **Then the same pipeline:** orientation interaction, lateral inhibition, stringing and parametric pruning, giving `flow_chains[n]`.
- **Relation to section 15:** the smear is the unoriented version of `flow_energy`. This field adds orientation and per-frame timing.
- **Check:**
    - `flow_energy` correlates with `cleaned_smears`.
    - On quiet pixels it sits at the noise level.
    - Where it is strong, its orientation agrees with the static orientation, which separates flow from leftover misregistration. Edge jitter moves across vessels, not along them.

25. Temporal signals per segment

a. Line mass `M_k(s, t)`. For static chain k, at each point s, integrate `temporal_frame` across the vessel (normal to θ, over ± `PAINT_WIDTH`·radius). The result is a kymograph image (s × t).

- Integrated density is roughly unchanged by defocus.
- In absorption imaging it is proportional to hemoglobin per unit length. Red cells are conserved at every junction, so this is the quantity to conserve.

b. Velocity, only where section 23 found visible flow.

- **Kymograph streaks:** red-cell streaks are lines in the kymograph, so run the static hypercolumn bank on it. The winning orientation φ gives speed `v = tan φ · fps` (px/s), and its sign the direction. Confidence is the streak's strength in noise units.
- **Cross-check with the 1-D continuity equation:** `v = −Σ ∂ₜM ∂ₛM / Σ (∂ₛM)²` over a window.

c. Slow signals, always computed.

- **Heart rate:** the spectral peak of the global mean density between 0.7 and 2.5 Hz.
- **Pulsatility per segment:** amplitude and phase of M at the heart rate and its first harmonic.
- **Vasomotion:** the band below 0.2 Hz, if the burst is long enough.
- Phase continuity along a vessel and across junction arms is a same-vessel cue. Arteries and veins separate by pulse phase and amplitude.

d. Focus trace and lucky windows.

- **Apparent half-width:** per chain, `w_k(t)` from scale selection on a 3-frame median, which averages out passing cells.
- **Relative blur:** `b_k(t) = √(w_k(t)² − min_t w_k²)`. A vessel's width is constant, so any change in apparent width is defocus.
- **Weights:** for velocity and flux, soft weights from `b_k(t)` over runs of at least `LUCKY_RUN` consecutive frames.
- **Co-focus:** arms that sharpen and blur together share a depth, which is a grouping cue.
- **Depth order:** if focus drifts monotonically over the burst, each chain's time of best focus orders the chains in depth.

Result: `segment_signals[n][k]` = dict(`M`, `times`, `velocity`, `velocity_confidence`, `direction`, `pulse_amplitude`, `pulse_phase`, `blur_trace`, `best_focus_time`).

Phase C — one field and one graph (section 26)

Agreement view (the combination the section-18 notes ask for). Compare the static and flow orientation scores pixel by pixel, at the same orientation:

Category	Meaning
both	perfused vessel
static only	no flow seen: not perfused, or flow too fast or too slow to see
flow only	faint or defocused vessel that the static field missed

Show the three in colour, with `static_chains`, `flow_chains` and their intersection overlaid.

Building the graph:

- **Combined field:** `U = max(U_static, U_flow)` per (x, y, θ), both in noise units, restrung as in section 22, giving `vessel_chains[n]`.
- **Graph:** nodes are chain ends and chain meetings; edges are the chain pieces between them.
- **Fuse split crossings.** A crossing often appears as two 3-way nodes joined by a short piece. Fuse them when they are closer than `max(3, 2.5 · W_thick / sin α)` and their outer arms pair straight across the piece. (W_thick is the thicker vessel's width; α is the crossing angle.)
- **Read the arms on a ring outside the junction core.** Inner radius is at least half the thick vessel's width plus 2 px; outer radius is several widths. Each arm records:
  - direction and blur-corrected width;
  - mean density and line mass;
  - velocity and direction (if measured);
  - pulse phase and blur trace.

  Measuring inside the core roughly doubles the angle error.

Phase D — junction reading (section 27)

D1. Baseline. vascnet's geometric rules:

- arms paired by straightness, with a relative margin;
- T test with amodal completion on the far side;
- Murray ratio;
- density additivity at the overlap;
- relocation to where the arm axes meet.

D2. Hypotheses are partitions of the arms into connected groups (Estrada et al., TPAMI 2015, generalized).

- **With flow directions:** each group needs at least one inflow and one outflow. Merges are allowed, because venules have real confluences.
- **Without directions:** every partition with at least two arms per group.

- **Example:** the 5-arm crossover-plus-bifurcation has six two-group hypotheses plus "unresolved".

D3. Score each partition as a sum of log terms, each in noise units:

- continuation within groups (turning angle, relatability);
- width and Murray consistency;
- blur continuity (co-focus) and pulse-phase continuity;
- density additivity at the overlap (two layers mean the densities add);
- flux balance per group: `Σ_in v·M̄ − Σ_out v·M̄`, time-resolved with transit lags when velocity is visible;
- temporal fingerprint: lagged correlation of M within groups, compared with across groups.

Decision rule:

- Prefer the most specific hypothesis that fits. A finer partition wins when it passes its tests, because the single "unresolved" group can never fail.
- Report "ambiguous" when two hypotheses tie within noise.
- Choose on even frames and verify on odd frames, so the scoring cannot talk itself into a balance.

D4. Network consistency.

- Orient edges by measured flow direction. With directions known, each junction's decision is independent of the others (Estrada's linear-time case).
- Edges without a reliable direction (stagnant or reversing flow) form small clusters, searched by flipping edge directions.

Result:

- `junctions[n]`: type, partition, confidence, arms, features.
- A JSON export in the shape of `Result.annotations()`.
- An overlay gallery ordered by confidence, for manual review.

Phase E — validation (alongside every phase)

E1. Synthetic bursts. Extend `vascnet.synth` with:

- discrete red-cell clusters advected through the network, with flux conserved at splits and merges;
- a depth per tree, and a focal plane that drifts during the burst;
- a 12-bit camera (saturation 4095), shot and read noise;
- the burst's frame rate and its gating gaps.

Truth: centerlines, junction types and partitions, velocities, depths.

E2. Metrics:

- centerline F1;
- junction detection F1;
- crossing recall and balanced type accuracy. Crossings are 20–25% of junctions, so plain accuracy flatters any classifier that defaults to "bifurcation".
- Estrada's parent and flow similarity;

- velocity error;
- the distribution of flux residuals.

E3. Real burst, without truth:

- repeatability across fixations;
- agreement between the static and flow fields;
- flux balance at clean Y junctions. It should centre on 0, and its spread is the flux noise used in D3.
- share of junctions resolved vs ambiguous;
- a manual spot check of the gallery.

Order of work

#	Build	Done when
1	Upstream changes (0)	`frame_times` exist; `temporal_frame` reproduces section 14's right-hand video
2	Sections 19–20	orientation scores and radius maps agree with section 18's best graph
3	Sections 21–22	the chains are found again across fixations at least as well as section 18's best picture, with fewer loose ends per 1000 px; a pruning plateau is chosen
4	Section 23	decision recorded: is flow visible, per width class, at this frame rate
5	E1 synthetic bursts	truth available for every later step
6	Section 24	`flow_chains` and the agreement view; flow energy at the noise level on quiet pixels
7	Section 25	heart rate found; per-segment pulse phase, blur traces and (if feasible) velocities
8	Section 26	one graph per fixation; split crossings fused
9	Section 27	junction table with partitions and confidence; on synthetic bursts, crossing recall above the D1 baseline

Starting values

Name	Value	Meaning
`N_ORIENT`	16	orientations per hypercolumn
`ELONGATION`	2.5	receptive-field length / width
`EDGE_SUPPRESSION`, `END_STOPPING`	1.0, 1.0	α, β of the complex unit
`THETA_CENTRE`, `THETA_SURROUND`	1, 3 bins	orientation-domain difference of Gaussians
`FLANK_OFFSET`	1–2 × radius	position of the inhibitory flanks
`LATERAL_STEPS`, `η`	4, 0.7	iterations and relaxation of section 21
`MODULATION`, `HALF_SAT`	0.5, 3	multiplicative facilitation
`T_LOW`, `T_HIGH`	2.0, 4.5	crest thresholds in noise units

Name	Value	Meaning
`GAP_MAX`	3 × radius	longest bridge between chain ends
`L_MIN`, `Z_MIN`	swept	pruning
`MAX_GAP_FRAMES`	1	kept-frame pairs used for differences
`LUCKY_RUN`	3 frames	shortest run for velocity

Risks and how each is caught

- **The frame rate may be too low to see red cells move.** Section 23 measures this first. Without visible motion, flow is detected by decorrelation, and timing comes from pulsatility and focus. Flux-based junction resolution then needs a faster burst.
- **Memory and time at full resolution.**
  - Wide scales run on reduced grids.
  - Flow statistics are streamed.
  - Per-frame signals are sampled only along chains.
  - vascnet's cached FFT bank is reused across frames.
- **Shading, flicker or misregistration can pass for flow.**
  - The flow field starts from `temporal_frame`, which has the shading removed.
  - Noise is measured on quiet pixels.
  - Flow must run along the static orientation, which the agreement view checks.
- **Holes from glare, saturation and frame edges.** Valid masks are carried through every layer, and no response is reported within a kernel radius of a hole.
- **A thin vessel can vanish inside a thick one.** Co-occurrence protection in section 21, and a dedicated check on synthetic crossings by width ratio and angle.